

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

Linrui Xu^a, Linrui Han^{b,c,*}

^a*University of Winnipeg, Winnipeg, Manitoba, Canada*

^b*Ministry of Education Laboratory of Philosophy and Social Sciences – The CUPL Data Law Lab, China University of Political Science and Law, Beijing, China*

^c*The Institute for Data Law, China University of Political Science and Law, Beijing, China*

Abstract

Recovering prior client-specific advice from a legal consultation archive is difficult because dense retrieval may rank a relevant question turn highly while omitting the lawyer’s answer—an *episode incompleteness* failure under same-domain interference. BIMS-LEGAL is a dual-store system for Chinese legal consultation logs: a fast episodic turn index with session identifiers and a slower BIRCH semantic store. Soft O2 performs turn-level session binding by giving siblings attenuated inherited scores (βs) rather than hard score copy. We evaluate exact replay as diagnostic and use paraphrase, follow-up, and advice-recall as primary non-exact channels. Primary Soft O2 contrasts use a shared turn-level index against dense FlatIP, hard hydration, and shuffled-*sid* controls; Soft O2 is strongest on non-exact channels where a partial episode hit can bind its answer sibling. Soft O2-C and gated Hybrid complement Soft O2 under a Mix cross-session protocol in which some gold evidence is held off-session; Soft O2 remains the default when query and gold already share a session identifier. We report graded nDCG alongside Answer Hit and Episode Completeness, with paired significance tests and a failure taxonomy separating complete, incomplete, and missed-session retrieval.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

*Corresponding author.

Email addresses: xu-l81@webmail.uwinnipeg.ca (Linrui Xu),
linrui_han@163.com (Linrui Han)

1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL is a dual-store Brain-Inspired Memory System for legal consultation archives, designed and evaluated in this study. The *systemic contribution* is a dual-store architecture that separates a fast episodic pathway (timestamped turn embeddings with speaker role and session identifier *sid*) from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) supplies an architectural analogy for this dual-pathway design, not an empirical claim about biological memory. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top- k —a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

36 The *algorithmic contributions* are soft binding operators that complete
 37 episodes without hard score copy:

- 38 • **Soft O2 (session soft binding).** When a turn is retrieved with score
 39 s , siblings sharing *sid* inherit $\max(s_{\nu}, \beta \cdot s)$, surfacing answer turns
 40 under question-hit / answer-miss while preserving rank competition
 41 among candidates.
- 42 • **Soft O2-C and gated Hybrid (cluster soft binding).** The same
 43 soft-inheritance rule applies within BIRCH clusters when gold evidence
 44 is not co-session with the query; Hybrid triggers cluster binding only
 45 on direct dense hits to avoid displacing same-session siblings.

46 Two **implementation and scale design operators** support rigorous large-
 47 scale shared-store experiments but are not standalone algorithmic contribu-
 48 tions:

- 49 • **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns,
 50 IVFPQ product quantization is under-trained and collapses Answer
 51 Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted
 52 sibling signals.
- 53 • **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes
 54 make archive construction tractable without changing retrieval seman-
 55 tics.

56 We ask four research questions:

- 57 • **RQ1.** Under same-domain interference, does episode incompleteness
 58 (question hit / answer miss) dominate full session miss?
- 59 • **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under
 60 measured index distortion, and what incremental gains does Soft O2
 61 provide over FlatIP and hard parent hydration?
- 62 • **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow
 63 (BIRCH) pathway effective? Does Soft O2-C help under standard same-
 64 session gold, and under a fair Mix of same-/cross-session gold?
- 65 • **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall
 66 channels, how large and session-tied are Soft O2 gains, and how do
 67 Soft O2 and Soft O2-C /Hybrid complement each other?

Contributions..

1. A dual-store memory architecture for Chinese legal consultation archives, separating fast episodic turn recovery from slower BIRCH semantic integration under same-domain interference.
2. Soft O2, a turn-level session soft-binding operator that addresses episode incompleteness without hard parent hydration or session-max score copy—the primary algorithmic claim before Soft O2-C.
3. A conservative three-candidate diagnostic of Soft O2 score competition using the gap ratio $\rho = s_d/s_h$; default $\beta=0.98$ is chosen by validation sweep, not by a closed-form optimum.
4. Turn-level lexical and fusion controls (BM25-turn, dense+BM25 RRF, cross-encoder) under the same evidence unit as Soft O2; joint episode / document-level indexes are a different input granularity and are not claimed as Soft O2 competitors here.
5. Soft O2-C and gated Hybrid under a Mix cross-session protocol, together with graded nDCG, a failure taxonomy, and paired significance reporting.
6. O1 exact FlatIP and O3 bulk-load consolidation as implementation operators enabling the shared-store ablations reported here.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL operators and the Soft O2 diagnostic model. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM

103 improve statutory and advisory generation Yue et al. (2023); Huang et al.
 104 (2023); Yue et al. (2024), but they still need a reliable memory index when
 105 prior advice must be recovered from a growing same-domain log.

106 Legal consultation archives add a further difficulty: high lexical overlap
 107 among topically adjacent matters, with the professionally useful evidence
 108 unit being the full question-answer episode. This setting motivates *dual-*
 109 *store* memory architectures that keep fast episodic turn retrieval separate
 110 from slower semantic clustering, and session-aware soft binding operators
 111 that recover answer turns under same-domain interference without collapsing
 112 rank competition among unrelated candidates.

113 2.2. Dual-store memory and episodic retrieval

114 Dual-store designs distinguish fast episodic encoding from slower seman-
 115 tic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving
 116 et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dia-
 117 logue agents extend context via hierarchical paging, summary banks, graphs,
 118 and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Ras-
 119 mussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang
 120 et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved
 121 units as flat passages rather than *multi-turn consultation episodes* whose
 122 answer turn may rank below an unrelated question turn sharing legal vo-
 123 cabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA
 124 over personal conversation logs Wu et al. (2024); Maharana et al. (2024),
 125 while broader memory-agent surveys emphasize persistence beyond the con-
 126 text window Wu et al. (2025); Tan et al. (2025). Recent conversational
 127 RAG work stresses reuse of historical search evidence across turns Mo et al.
 128 (2026). BIMS-LEGAL positions its dual-store contribution at this intersec-
 129 tion: episodic turn embeddings with *sid* for same-session recovery (Soft O2),
 130 and BIRCH centroids with Soft O2-C for cross-session recovery when query
 131 and gold evidence occupy different sessions but share thematic structure
 132 Zhang et al. (1996). Soft O2 directly addresses the episodic retrieval chal-
 133 lenge that parent-document hydration and session-max aggregation handle
 134 via hard score copy: under dense same-domain interference, a question turn
 135 can dominate top- k while its answer sibling remains absent, producing fluent
 136 yet ungrounded downstream generation unless siblings enter under attenu-
 137 ated inherited scores.

138 2.3. Session-aware retrieval

139 Parent-document hydration, session aggregation, and joint session index-
 140 ing are standard session-aware mechanisms for attaching sibling context to a
 141 dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps
 142 ranking competition among candidates. Parent-document and hierarchical
 143 retrieval similarly attach child evidence to a parent unit Callan et al. (1994);
 144 Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2
 145 differs by using attenuated turn-level score propagation rather than hard hy-
 146 dration or atomic session documents. Lexical BM25 Robertson and Zaragoza
 147 (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Recipro-
 148 cal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira
 149 et al. (2019); Chen et al. (2024) provide complementary controls outside ses-
 150 sion membership. Exact flat indexes and product-quantized ANN backends
 151 Johnson et al. (2019) further affect whether sibling binding is applied over
 152 faithful turn scores. Table 1 summarizes the session-structure contrasts used
 153 in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

154 *On the reported corpora, session-max rankings coincide with hard parent hydration
 155 under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2
 156 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH path-
 157 way.

158 We reuse CLS vocabulary as an architectural analogy for BIMS-LEGAL:
 159 turn embeddings with timestamps, roles, and session identifiers as episodic
 160 elements; Soft O2 as episode binding on the episodic pathway; BIRCH cen-
 161 troids with Soft O2-C as semantic integration on the slow pathway; and
 162 shared-store distractors as interference. The dual-store split is the systemic
 163 design; Soft O2 and Soft O2-C are the algorithmic mechanisms evaluated on
 164 Chinese legal consultation archives.

165 3. Method: BIMS-LEGAL

166 3.1. Architecture overview

167 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
168 tary episodic and semantic stores over the same consultation archive.

169 **Definition 1** (Episodic and semantic elements). *An episodic element is*
170 $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, *where* $\mathbf{v}_i \in \mathbb{R}^d$ *is an embedding,* t_i *a timestamp,* c_i *conversational context (including speaker role),* ϕ_i *surface features, and* sid_i *the consultation session identifier. A semantic element is* $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ *with centroid* $\mathbf{c}_j$ *, member set* C_j *, label* κ_j *, and formation statistics* τ_j *. Write* $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$ *for session siblings and* $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$ *for BIRCH-cluster siblings.*

176 **Definition 2** (Dual-store retrieval map). *Given query* q *with embedding*
177 $\mathbf{v}_q$, *base episodic retrieval returns a scored candidate set* $\mathcal{R}_0(q) = \{(t, s_t)\}$ *under Eq. (1). Soft O2 produces* $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ *(Sec-*
178 *tion 3.3). Soft O2-C produces* $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ *(Sec-*
179 *tion 3.5). Gated Hybrid composes session binding then cross-session cluster*
180 *binding (Algorithm 2). The returned list is the top-k of the bound set under*
181 *the rewritten scores.*

183 3.2. Base scoring and implementation operators (O1, O3)

184 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
185 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

186 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

187 with Δ_t the elapsed time between query time and the turn timestamp. Unless
188 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
189 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
190 and do not replace Eq. (1).

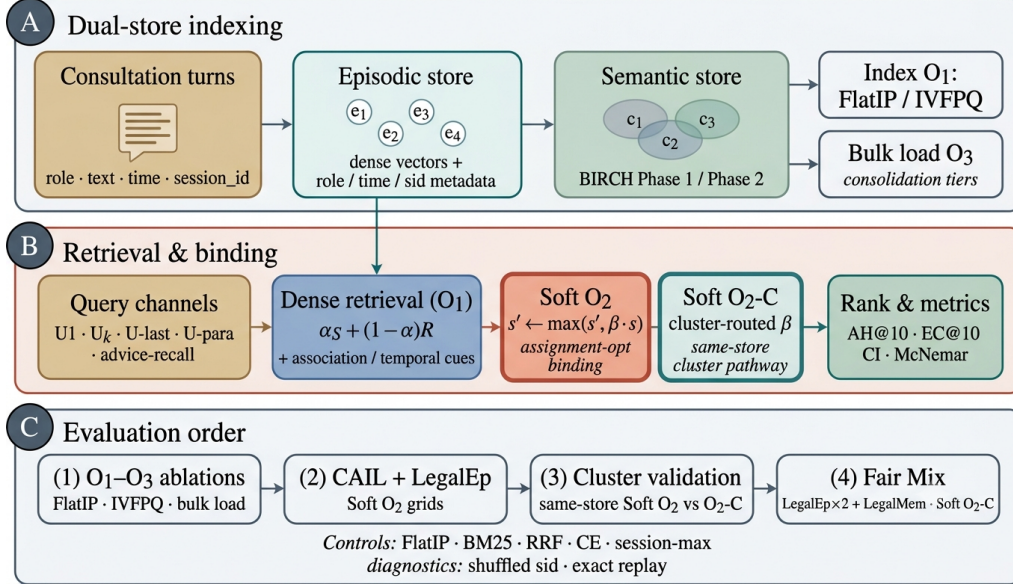


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load consolidation). (B) Query-time dense retrieval under O1 FlatIP with Soft O2 session binding and Soft O2-C cluster binding (algorithmic core); O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, CAIL /LegalEp Soft O2 grids, same-store cluster validation, and fair Mix Soft O2-C.

191 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 192 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 193 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 194 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 195 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 196 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 197 engineering prerequisite for rigorous Soft O2 evaluation, not a standalone
 198 retrieval algorithm.

199 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 200 insert update ($e_i \mapsto \text{BIRCH}(e_i); \text{Flush}$) by a deferred operator **Finalize** =
 201 $\text{BIRCH}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic writes. O3 does
 202 not change retrieval semantics; it makes $M \geq 400$ shared-store experiments
 203 runnable (≈ 11 min wall-clock for $M=400$; amortized ≈ 0.825 s/turn over
 204 ~ 800 turns). Quality-latency curves to $M=6400$ appear in Section 5.6.

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

205 *3.3. Soft O2: session soft binding*

206 *Motivation..* Under same-domain interference, a retrieved question turn can
207 score highly while its answer sibling remains outside top- k —episode incom-
208 pleteness compounded by IVFPQ score collapse and context fragmentation.
209 Soft O2 completes episodes without hard score copy.

210 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
211 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

212 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
213 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
214 *each member score by $\max_{t \in sid} s_t$.*

215 Algorithm 1 states the deployed Soft O2 procedure (expand + complete-
216 ness truncate), matching the public implementation.

217 *3.4. Soft O2 attenuation as diagnostic score competition*

218 The attenuation β controls how strongly a retrieved turn promotes its
219 session siblings. We treat the following as a *diagnostic* model, not a proof

of a universal optimum. In particular, a zero-margin hinge objective with $\beta \in (0, 1]$ does not identify a meaningful quantile solution: when the rank margin is zero, the rank term is inactive for all $\beta < 1$, so its weight cannot determine a closed-form optimum.

Definition 4 (Three-candidate diagnostic model). *For a query whose dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be the gold answer sibling and let d be the strongest non-session distractor in a wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured distractor, conditions based on ρ are conservative sufficient conditions for beating that distractor, not exact top- k boundary conditions for every competitor.

The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$, where $s'_a = \beta s_h$.

Proposition 1 (Feasible attenuation band). *Under Definition 4, assume the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$, any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$ already, rank preservation depends on the observed sibling score rather than on β alone.*

Proof. If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e. $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when $s_h > 0$. $\square$

Exploratory soft-rank diagnostic. We also minimize an exploratory soft pairwise diagnostic on a validation grid,

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

with logistic σ , temperature τ , and rank weight γ . This visualization is useful for studying the availability–ranking trade-off, but it depends on the diagnostic pool, query channel, and chosen hyperparameters. Default $\beta=0.98$

Table 2: Diagnostic gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). These are calibration diagnostics, not derivations of an optimal β .

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

is selected from the sweep in Table A.4, where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band.

Fair controls include hard parent hydration, session-max aggregation, and shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max coincide under full-sibling expansion; we report a single hard-expansion baseline. Indexes that concatenate an entire question–answer episode into one retrieval unit (BM25-joint or dense joint-episode documents) change the input granularity and are therefore outside the Soft O2 turn-level protocol; we do not report them as Soft O2 competitors.

3.5. Semantic consolidation and Soft O2-C

Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang et al. (1996).

Definition 5 (BIRCH dual-phase consolidation). *Let $\{\mathbf{c}_j\}$ be current centroids and θ_H, θ_M the assign /merge thresholds. **Phase 1 (assign)**. For a new embedding $\mathbf{v}$, set $j^* = \arg \max_j S(\mathbf{v}, \mathbf{c}_j)$; if $S(\mathbf{v}, \mathbf{c}_{j^*}) \geq \theta_H$ then $C_{j^*} \leftarrow C_{j^*} \cup \{\mathbf{v}\}$ and refresh $\mathbf{c}_{j^*}$, else spawn a new cluster. **Phase 2 (merge)**. Periodically, merge pairs with $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ and purge near-duplicates above a redundancy threshold. Under O3, both phases run inside Finalize. Each turn t receives a cluster id cid_t , inducing $\text{Clus}(t)$ in Definition 1.*

Definition 6 (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default $\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session membership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large thematic clusters cannot flood top- k .*

Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings relative to sessions already covered by dense /Soft O2 hits, and triggers cluster binding only on direct dense hits. This prevents thematic confusers

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$   $\triangleright$  session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$   $\triangleright$  direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

276 from displacing same-session siblings already recovered by Soft O2. Soft O2
277 and Soft O2-C are complementary: Soft O2 is the default binder when query
278 and gold share sid ; Soft O2-C is required when gold evidence is held in a
279 different session but the same cluster.

280 4. Corpora and Evaluation Protocol

281 Figure 2 sketches the shared-corpus design used throughout the legal eval-
282 uation. Gold needles and same-domain distractors occupy one store. Each
283 query probes that shared archive under fixed interference conditions. Pri-
284 vately rebuilt haystacks are avoided so that system comparisons isolate the
285 retrieval operator.

286 4.1. Datasets and query channels

287 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
288 Gold needles are authentic preliminary and final dialogues, with conversation
289 turns completed by label turns where applicable. Query channels cover the
290 first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user
291 turn (U-last), and constrained paraphrases (U-pa). Evidence defaults to
292 assistant turns in the target session.

293 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
294 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
295 a natural (question, answer) pair with session identifier sid . Reconstruction

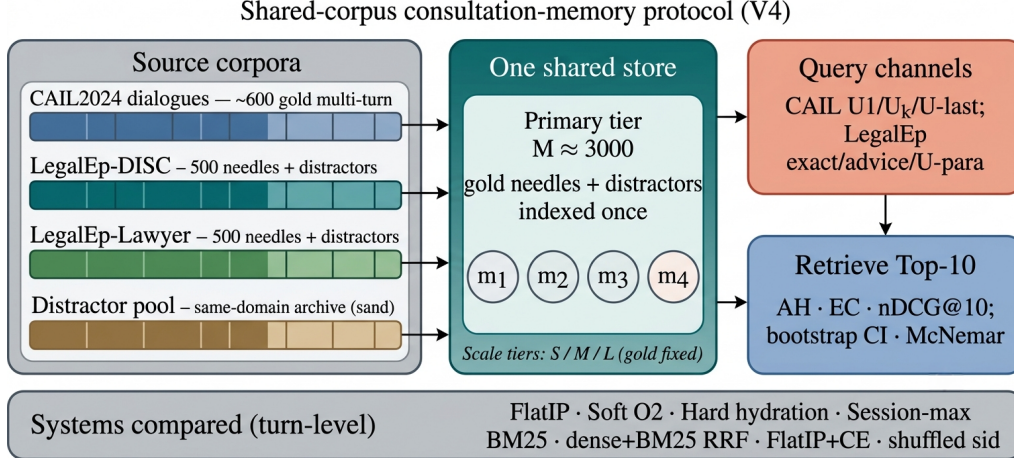


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

296 applies length bounds and legal-content filters, excludes exam prompts, re-
 297 moves near-duplicates, separates needles from distractors under a fixed seed,
 298 and retains an audit subsample. After filtering, each LegalEp corpus retains
 299 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
 300 tier).

301 *Query-channel policy (addressing exact-match shortcuts)*. Exact replay uses
 302 the stored user question and is reported only as a *diagnostic* upper bound
 303 on surface-form recovery. Primary claims use paraphrase, Uk-followup /U-
 304 last, and advice-recall queries whose surface form is not identical to the
 305 stored question text; paraphrase and advice-recall templates are generated
 306 under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL
 307 multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C
 308 contrasts and for Mix reconstructions.

309 Archive size is varied with gold needles held fixed. Unless stated otherwise,
 310 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,
 311 $S=300$.

312 4.2. Metrics and baselines

313 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
 314 and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is

315 binary: whether a gold answer turn appears in the top- k . EC is the mean
 316 coverage of gold turns within the target session. Session Hit@ k is reported
 317 only for diagnosis. For nDCG we use fixed graded relevance over the full
 318 gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all
 319 other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all
 320 gold-relevant turns in the target session and truncated at k , so every system
 321 on a query shares the same IDCG. Answer Hit and Episode Completeness
 322 are the primary Soft O2 claims; nDCG uses the fixed gold-session IDCG
 323 above so that systems remain comparable on each query. Primary contrasts
 324 use bootstrap 95% CIs and McNemar mid- p tests on paired per-query hits;
 325 where families of related hypotheses are interpreted jointly, we also report
 326 Holm-corrected p -values (Table A.2).

327 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
 328 soft sibling score inheritance on the episodic pathway. Soft O2-C applies
 329 the analogous rule within BIRCH clusters. Hard parent hydration expands
 330 a hit to all siblings by copying the trigger score. Lexical controls include
 331 BM25 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the
 332 same turn-level evidence unit. We do *not* include BM25-joint or dense joint-
 333 episode indexes in Soft O2 primary comparisons: those methods retrieve pre-
 334 concatenated episode documents and therefore answer a different retrieval
 335 problem. A cross-encoder (CE) control reranks the top-30 FlatIP candidates
 336 with BAAI/bge-reranker-v2-m3. Negative and sensitivity controls include
 337 shuffled *sid*, a β sweep, and IVFPQ as a quantization reference.

338 4.3. Fair Mix protocol for Soft O2-C

339 On fully same-session LegalMem grids, Soft O2 already recovers gold
 340 answers that share *sid* with the query, so Soft O2-C has little exclusive head-
 341 room and may inject thematic confusers. We therefore reconstruct the *same*
 342 LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes
 343 are split into query-side S_a and evidence-side S_b with distinct session identi-
 344 fiers, while 30% remain intact. Distractors are unchanged. All operators—
 345 FlatIP, Soft O2, Soft O2-C, and gated Hybrid—share unrestricted dense re-
 346 trieval on this store; we report overall AH and stratum AH on `cross_session`
 347 vs `same_session` queries.

348 *Cluster co-membership under Mix..* After BIRCH consolidation on the Mix
 349 store, each cross-session S_a/S_b split pair is *glued* into one cluster identifier
 350 so that Soft O2-C can be evaluated when thematic co-membership is present.

351 Glue uses split metadata from the evaluation construction only; it does not
 352 inject labels at query time. If Soft O2-C still fails to recover cross-session
 353 gold under that condition, the failure is attributable to binding rather than
 354 accidental cluster separation. Natural same-cluster rates without glue are
 355 lower and are noted in Section 6.3. Soft O2’s session membership constraint
 356 is unchanged: glue affects only whether Soft O2-C has a cluster in which to
 357 bind. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split
 358 Uk-followup queries. External LongMemEval /LoCoMo numbers from prior
 359 retrieval runs on this codebase (QA correctness 70.7% / 68.2% on released
 360 test splits) are cited only as long-horizon memory context Wu et al. (2024);
 361 Maharana et al. (2024), not as Soft O2-C evidence on legal corpora.

362 5. Experiments

363 Unless noted, the embedding model is `nomic-embed-text-v2-moe` Nuss-
 364 baum et al. (2025), $k=10$, seed 42. We report primary contrasts in figures
 365 and keep full numeric grids in the appendix. Result files and scripts are re-
 366 leased with the accompanying repository so that each table maps to a public
 367 JSON artifact.

368 5.1. Implementation operators and Soft O2 ablation

369 Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA
 370 ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone
 371 yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirm-
 372 ing that exact indexing is a necessary implementation prerequisite but insuffi-
 373 cient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283
 374 over baseline) and Episode Completeness to 0.888 / 0.840—the primary al-
 375 gorithmic gain. O3 is enabled for all configurations in this ladder as a con-
 376 struction prerequisite; it does not alter retrieval scores.

377 5.2. RQ1 failure taxonomy

378 We classify each query into *complete* (answer evidence in top- k), *incom-*
 379 *plete* (target session partially retrieved but answer missing), or *session miss*
 380 (no target-session turn in top- k). Table 4 contrasts FlatIP, Soft O2, hard
 381 hydration, and shuffled *sid* on primary non-exact channels. Soft O2 converts
 382 incompleteness into complete retrieval while session-miss stays nearly fixed;
 383 shuffled *sid* restores high incompleteness, so the gain is session-tied rather
 384 than generic score inflation. Hard hydration can also drive incompleteness

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M \approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

near zero by copying trigger scores, but it is not Soft O2’s binding rule and can over-promote siblings (Section 5.4).

5.3. Turn-level Soft O2 controls

Primary Soft O2 claims use a shared turn-level index. Table 5 reports Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel; shuffled *sid* collapses toward or below FlatIP; hard hydration is competitive on CAIL AH but Soft O2 remains stronger on LegalEp paraphrase /advice and retains higher EC in the full grids (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8): Soft O2 exceeds it on CAIL Uk /U-last (0.698/0.762 vs 0.482/0.413). Joint episode / document-level indexes change evidence granularity and are not Soft O2 protocol competitors.

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200), tying the improvement to correct episode membership (RQ1–RQ2). Hard expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a rank-preserving binder: siblings enter under attenuated scores, and episode coverage remains higher than under hard score copy. Table 6 shows the Soft O2>FlatIP ordering under graded nDCG with gold-session IDCG,

Table 4: RQ1 failure taxonomy ($k=10$). Boldface: best complete / lowest incomplete within each corpus-channel block among FlatIP, Soft O2, and shuffled *sid* (hard hydration shown as a score-copy envelope). Soft O2 cuts incompleteness without raising session miss; shuffled *sid* undoes the gain.

Corpus / channel	System	Complete	Incomplete	Session miss
DISC / u-param	FlatIP	70.8%	26.8%	1.6%
	Soft O2	89.1%	8.7%	1.8%
	Hard hydr.	97.2%	0.0%	2.8%
	Shuffled O2	66.4%	31.4%	2.2%
DISC / advice	FlatIP	48.0%	17.0%	35.0%
	Soft O2	60.8%	3.8%	35.4%
	Hard hydr.	65.0%	0.0%	35.0%
	Shuffled O2	45.6%	19.2%	35.2%
Lawyer / u-param	FlatIP	71.5%	25.3%	0.8%
	Soft O2	94.0%	3.8%	1.6%
	Hard hydr.	98.0%	0.0%	2.0%
	Shuffled O2	72.9%	25.5%	1.6%
CAIL / Uk	FlatIP	45.8%	54.0%	0.2%
	Soft O2	92.8%	7.0%	0.2%
	Hard hydr.	98.3%	1.5%	0.2%
	Shuffled O2	34.7%	65.2%	0.2%

while incompleteness falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

Figure 4 reports LegalEp transfer on the same AH grids. On LegalEp-DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already recoverable for many queries—consistent with treating exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNemar mid- $p=0.024^*$). Paraphrase yields the clearest LegalEp gains: Soft O2 reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, exceeding hard expansion on both corpora (mid- $p<0.001^{**}$ on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid- $p<0.001^{**}$), while shuffled *sid* falls toward or below FlatIP. Graded nDCG mirrors the Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is the session-binding claim; shuffled *sid* tests membership dependence; hard hydration is score-copy expansion.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.570	0.788	0.802	0.452
CAIL / Uk	0.270	0.698	0.720	0.230
CAIL / U-last	0.245	0.762	0.790	0.200
LegalEp-DISC / u-param	0.547	0.648	0.592	0.469
LegalEp-DISC / advice	0.342	0.428	0.402	0.310
LegalEp-Lawyer / u-param	0.568	0.759	0.596	0.592
LegalEp-Lawyer / advice	0.332	0.472	0.394	0.348

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

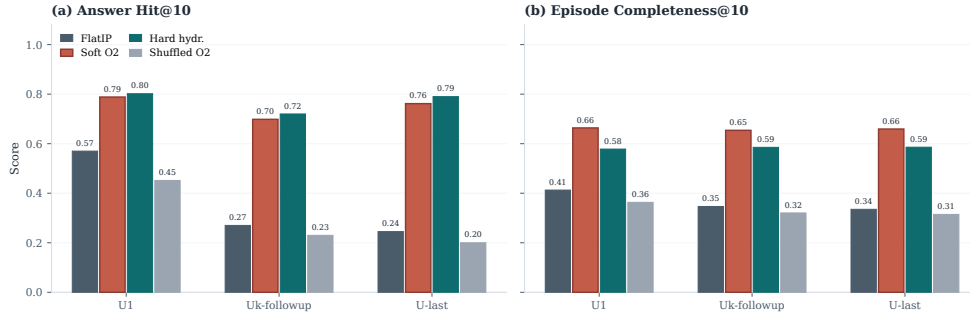


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

423 Cross-encoder reranking is a strong neural control at the same turn-
 424 level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with
 425 **bge-reranker-v2-m3**. Table 7 shows Soft O2 ahead of FlatIP+CE on CAIL
 426 multi-turn channels and on LegalEp-Lawyer exact, while FlatIP+CE leads on
 427 LegalEp-DISC exact—consistent with Soft O2’s largest gains arising where a
 428 partial episode hit can bind its answer sibling rather than where surface-form
 429 overlap already recovers both turns.

430 5.5. Soft O2-C and gated Hybrid under Mix

431 When gold answers already share *sid* with the query, Soft O2 is the
 432 natural operator and cluster binding is unnecessary. The complementary
 433 question is whether Soft O2-C helps when evidence is deliberately placed off-

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	0.793	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	0.791	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	0.759	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	0.524	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	0.844	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	0.582	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

session. Table 8 uses the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated Hybrid sharing unrestricted dense retrieval. On the cross-session stratum, Soft O2-C improves over Soft O2 on LegalEp-Lawyer (0.491 vs 0.446) and LegalMem-MT (0.537 vs 0.503), showing that cluster soft binding can recover split-episode gold that session membership alone cannot surface. Gated Hybrid Soft O2+Soft O2-C turns this mechanism into an overall gain: LegalEp-Lawyer AH rises to 0.534 from Soft O2 0.496 (mid- $p=0.010^*$), and LegalMem-MT reaches 0.646 vs Soft O2 0.618, while most same-session hits are preserved (0.560 / 0.867). Ungated Soft O2-C can raise cross-session hits yet displace intact same-session evidence, which is why Hybrid triggers cluster binding only on direct dense hits. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led and are reported in Appendix Appendix A (Table A.1).

McNemar mid- p on paired Answer Hit; * $p<0.05$, ** $p<0.01$. Cluster co-membership of Sa/Sb pairs follows the Mix protocol in Section 4.3.

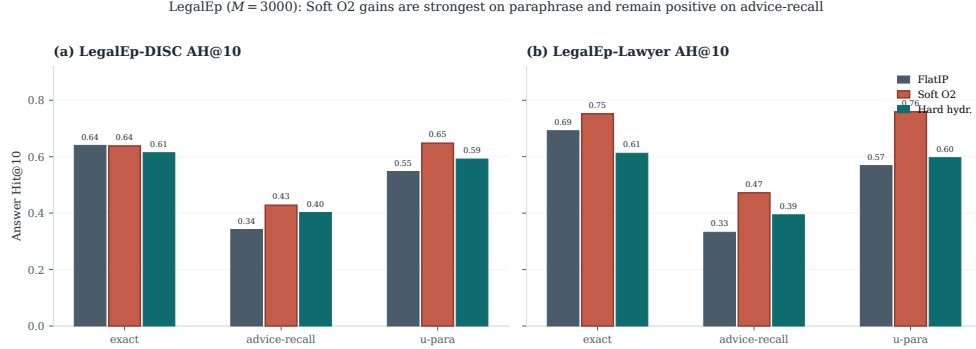


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Cross-encoder control (AH@10). FlatIP+CE reranks the top-30 FlatIP candidates with `bge-reranker-v2-m3`. Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on LegalEp-DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`) links retrieval quality to downstream answerability under a *single* retrieval configuration (Soft O2); we do not claim causal superiority over FlatIP in generation without paired downstream controls. Full protocol and stratified results appear in Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

Table 8: Mix evaluation of Soft O2-C and gated Hybrid (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p < 0.05$, ** $p < 0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	0.534*	0.523	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	0.646	0.551	0.867	0.1837

6. Discussion

6.1. Findings and scope

RQ1–RQ2 show that episode incompleteness is a substantial recurrent FlatIP failure mode under same-domain interference, and that Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity as an implementation prerequisite; Soft O2 supplies the primary algorithmic gain through soft sibling inheritance; O3 enables archive construction at scale. The diagnostic gap-ratio analysis (Section 3.4) explains why β should remain close to but below unity, and why the validation sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. LegalEp paraphrase and advice-recall—not exact replay—carry the primary Soft O2 transfer claims. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). RQ3–RQ4 follow Soft O2 $\rightarrow$ Soft O2-C: under Mix, gated Hybrid recovers additional cross-session gold (Table 8), while Soft O2 remains the default when gold already shares *sid* (Table A.1). Sparse and fusion turn-level baselines situate Soft O2 among standard IR alternatives

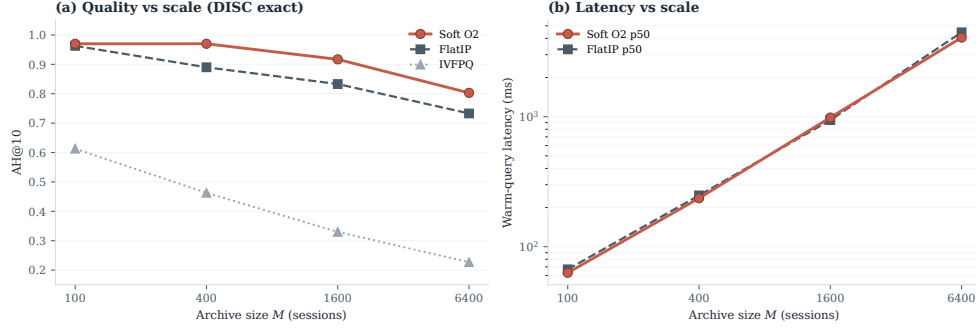


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019);
coarser joint-episode indexes were not evaluated as Soft O2 competitors.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder `legal_qa_n270`). Table 9 summarizes pooled Evidence-Answerability Rate (EAR) and stratified independent-judge scores. When Soft O2 retrieves complete episodes—both question and answer turns in top- k —the generator receives grounded prior advice and judged answerability rises in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60/\text{corpus}$) yields Cohen’s $\kappa=0.71$ / 0.68 and separates retrieval from generation failures (retrieval-failure share 0.283 / 0.317). Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for

507 deployment. AH and EAR measure evidence availability and judged answer-
508 ability respectively; statutory correctness, citation validity, and professional
509 liability lie outside this evaluation.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). EAR: Evidence-Answerability Rate (judge score ≥ 0.7). Stratified judge scores: $n=90$ /corpus across exact, paraphrase, and follow-up protocols.

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

510 6.3. Limitations

511 LegalEp episodes are consultation pairs; multi-turn authenticity remains
512 reserved for CAIL. The Mix protocol co-locates Sa/Sb pairs so Soft O2-C
513 can be tested when thematic co-membership is present (Section 4.3); un-
514 supervised BIRCH without that construction step would not receive split-
515 pair metadata at build time. Soft O2 depends on correct session identifiers;
516 Soft O2-C depends on cluster quality. BM25-joint and dense joint-episode
517 document indexes were not included in Soft O2 primary comparisons because
518 they change the retrieval evidence unit from turns to concatenated episodes;
519 evaluating those coarser designs under a matched online-update protocol is
520 left to future work. The β diagnostic is semi-empirical: it assumes a dominant
521 distractor and a well-defined hit score, and the fitted ρ uses a stored user-turn
522 embedding as an exact-channel query proxy; paraphrase /advice-recall gap
523 laws may shift the feasible band slightly but leave the high- β operating region
524 unchanged. LLM paraphrases need human audit. Statutory correctness, cita-
525 tion validity, and professional liability remain unevaluated. Multiple channels
526 inflate comparison multiplicity; we therefore report Holm-corrected p -values
527 for the primary Soft O2 vs FlatIP family (Table A.2) alongside uncorrected
528 McNemar mid- p tests.

529 *Latency and deployment scope..* Scale curves cover $M \in$
530 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.3). At
531 $M=6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms
532 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL
533 should therefore be positioned as **high-precision offline or quasi-**
534 **real-time consultation memory recovery**—e.g., pre-meeting case

review, audit replay, or batch RAG over a firm’s advice archive—not as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly 10^5 sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

7. Conclusion

BIMS-LEGAL targets consultation-memory retrieval under same-domain interference in Chinese legal archives through a dual-store design: an episodic turn index paired with a BIRCH semantic store. Soft O2 session soft binding is the primary algorithmic contribution; Soft O2-C with gated Hybrid extends soft binding to BIRCH clusters under Mix reconstructions in which some gold evidence is held off-session; Soft O2 attenuation β is validated on a diagnostic gap-ratio model; and O1 exact indexing with O3 bulk-load consolidation are implementation operators that enable the shared-store ablations reported here. Under a shared-corpus protocol spanning CAIL multi-turn dialogues and rebuilt LegalEp episode corpora, Soft O2 improves Answer Hit over FlatIP on multi-turn, paraphrase, and advice-recall channels with collapsing shuffled-*sid* controls, yields higher episode completeness than hard expansion on CAIL, and raises graded nDCG while reducing incompleteness. Under Mix reconstructions, gated Hybrid Soft O2+Soft O2-C recovers additional cross-session gold while Soft O2 remains the default on intact same-session gold. End-to-end QA audit links complete episode retrieval to judged answerability and low wrong-session contamination in downstream generation under Soft O2 retrieval. Sparse, fusion, and cross-encoder baselines situate Soft O2 among standard turn-level IR alternatives; joint episode / document-level indexes are a different input granularity and are not Soft O2 competitors in this protocol.

Declarations

Funding

Computational resources were provided by the Data Law Laboratory, China University of Political Science and Law.

568 *Conflict of interest*

569 The authors declare no competing interests.

570 *Ethics approval*

571 This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory retrieval.

574 *Data availability*

575 Code, processed evaluation corpora, and primary result files are publicly available at <https://github.com/Kilimajaro/soft-episode-binding-legal-memory>. Upstream legal corpora: CAIL2024 consultation tracks; Hugging Face ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_data.

580 *Code availability*

581 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load, Soft O2-C), LegalMem/LegalEp pipelines, Mix builders, and table/figure scripts are released in the same repository with JSON artifacts for each main table.

585 *Author contributions*

586 L.X. (University of Winnipeg) designed BIMS-LEGAL and the Soft O2 /Soft O2-C evaluation, implemented the system, ran the experiments, and wrote the manuscript. L.H. (corresponding author; CUPL Data Law Lab and Institute for Data Law) provided methodology guidance, revision, and supervision.

591 **References**

- 592 Ashley, K.D., 2017. Artificial Intelligence and Legal Analytics: New Tools
593 for Law Practice in the Digital Age. Cambridge University Press.
- 594 Askari, A., Aliannejadi, M., Abolghasemi, A., Kanoulas, E., Verberne, S.,
595 2024. Answer retrieval in legal community question answering, in: Ad-
596 vances in Information Retrieval (ECIR), Springer. pp. 477–485.
- 597 Atkinson, R.C., Shiffrin, R.M., 1971. The control of short-term memory.
598 Scientific American 225, 82–91.
- 599 Callan, J.P., Lu, Z., Xu, S., 1994. Passage-level evidence in document re-
600 trieval, in: Proceedings of the 17th Annual International ACM SIGIR
601 Conference on Research and Development in Information Retrieval, pp.
602 302–310.
- 603 Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z., 2024. M3-
604 embedding: Multi-linguality, multi-functionality, multi-granularity text
605 embeddings through self-knowledge distillation, in: Findings of the associ-
606 ation for computational linguistics: ACL 2024, pp. 2318–2335.
- 607 Cormack, G.V., Clarke, C.L.A., Buettcher, S., 2009. Reciprocal rank fusion
608 outperforms condorcet and individual rank learning methods, in: Proceed-
609 ings of the 32nd International ACM SIGIR Conference on Research and
610 Development in Information Retrieval, pp. 758–759.
- 611 Dai, Z., Xiong, C., Liu, Z., Callan, J., 2019. Context-aware sentence/passage
612 encoding for answering factoid questions, in: Proceedings of the 2019 Con-
613 ference on Empirical Methods in Natural Language Processing and the
614 9th International Joint Conference on Natural Language Processing, pp.
615 241–251.
- 616 Gutiérrez, B.J., Shu, Y., Gu, Y., Yasunaga, M., Su, Y., 2024. Hipporag:
617 Neurobiologically inspired long-term memory for large language models.
618 Advances in neural information processing systems 37, 59532–59569.
- 619 Huang, Q., Tao, M., Zhang, C., An, Z., Jiang, C., Chen, Z., Wu, Z., Feng,
620 Y., 2023. Lawyer llama technical report. [arXiv:2305.15062](https://arxiv.org/abs/2305.15062).
- 621 Johnson, J., Douze, M., Jégou, H., 2019. Billion-scale similarity search with
622 GPUs. IEEE Transactions on Big Data 7, 535–547.

- 623 Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen,
624 D., Yih, W.t., 2020. Dense passage retrieval for open-domain question
625 answering, in: Proceedings of the 2020 Conference on Empirical Methods
626 in Natural Language Processing (EMNLP), pp. 6769–6781.
- 627 Kumaran, D., Hassabis, D., McClelland, J.L., 2016. What learning systems
628 do intelligent agents need? complementary learning systems theory up-
629 dated. Trends in cognitive sciences 20, 512–534.
- 630 Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küt-
631 tler, H., Lewis, M., Yih, W.t., Rocktäschel, T., et al., 2020. Retrieval-
632 augmented generation for knowledge-intensive nlp tasks. Advances in neural
633 information processing systems 33, 9459–9474.
- 634 Liu, B., Wu, Y., Zhang, F., Liu, Y., Wang, Z., Li, C., Zhang, M., Ma, S., 2022.
635 Query generation and buffer mechanism: Towards a better conversational
636 agent for legal case retrieval. Information Processing & Management 59,
637 103051.
- 638 Louis, A., van Dijck, G., Spanakis, G., 2024. Interpretable long-form legal
639 question answering with retrieval-augmented large language models, in:
640 Proceedings of the AAAI Conference on Artificial Intelligence, pp. 22266–
641 22275.
- 642 Maharana, A., Lee, D.H., Tulyakov, S., Bansal, M., Barbieri, F., Fang, Y.,
643 2024. Evaluating very long-term conversational memory of llm agents, in:
644 Proceedings of the 62nd Annual Meeting of the Association for Computa-
645 tional Linguistics (Volume 1: Long Papers), pp. 13851–13870.
- 646 Martinez-Gil, J., 2023. A survey on legal question–answering systems. Com-
647 puter Science Review 48, 100552.
- 648 McClelland, J.L., McNaughton, B.L., O’Reilly, R.C., 1995. Why there are
649 complementary learning systems in the hippocampus and neocortex: in-
650 sights from the successes and failures of connectionist models of learning
651 and memory. Psychological review 102, 419.
- 652 Mo, F., Gao, Y., Wu, Z., Liu, X., Chen, P., Li, Z., Wang, Z., Li, X., Jiang,
653 M., Nie, J.Y., 2026. Leveraging historical information to boost retrieval-
654 augmented generation in conversations. Information Processing & Man-
655 agement 63, 104449.

- 656 Modarressi, A., Imani, A., Fayyaz, M., Schütze, H., 2023. Ret-llm: Towards
657 a general read-write memory for large language models. arXiv preprint
658 arXiv:2305.14322 .
- 659 Nogueira, R., Yang, W., Cho, K., Lin, J., 2019. Multi-stage document rank-
660 ing with BERT. arXiv:1910.14497.
- 661 Nussbaum, Z., et al., 2025. Nomic embed multimodal: Interleaved text, im-
662 age, and datasets for general-purpose vision-language embeddings. Model
663 card: **nomic-embed-text-v2-moe** (Nomic AI). Embedding model used for
664 turn encoding in this work.
- 665 Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S.G., Stoica, I., Gonzalez,
666 J.E., 2023. Memgpt: Towards llms as operating systems. arXiv preprint
667 arXiv:2310.08560 .
- 668 Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., Chalef, D., 2025. Zep:
669 A temporal knowledge graph architecture for agent memory. arXiv preprint
670 arXiv:2501.13956 .
- 671 Robertson, S., Zaragoza, H., 2009. The probabilistic relevance framework:
672 BM25 and beyond. Foundations and Trends in Information Retrieval 3,
673 333–389.
- 674 Tan, H., Zhang, Z., Ma, C., Chen, X., Dai, Q., Dong, Z., 2025. Mem-
675 bench: Towards more comprehensive evaluation on the memory of llm-
676 based agents, in: Findings of the Association for Computational Linguis-
677 tics: ACL 2025, pp. 19336–19352.
- 678 Tulving, E., et al., 1972. Episodic and semantic memory. Organization of
679 memory 1, 1.
- 680 Wang, Y., Gao, Y., Chen, X., Jiang, H., Li, S., Yang, J., Yin, Q., Li, Z.,
681 Li, X., Yin, B., et al., 2024. Memoryllm: Towards self-updatable large
682 language models. arXiv preprint arXiv:2402.04624 .
- 683 Wu, D., Wang, H., Yu, W., Zhang, Y., Chang, K.W., Yu, D., 2024. Long-
684 memeval: Benchmarking chat assistants on long-term interactive memory.
685 arXiv preprint arXiv:2410.10813 .

686 Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R.,
687 Liu, Y., 2025. From human memory to ai memory: A survey on memory
688 mechanisms in the era of llms. [arXiv:2504.15965](#).

689 Yue, S., Chen, W., Wang, S., Li, B., Shen, C., Liu, S., Zhou, Y., Xiao, Y.,
690 Yun, S., Huang, X., Wei, Z., 2023. Disc-lawllm: Fine-tuning large language
691 models for intelligent legal services. [arXiv:2309.11325](#).

692 Yue, S., Liu, S., Zhou, Y., Shen, C., Wang, S., Xiao, Y., Li, B., Yun, S.,
693 Shen, X., Chen, W., et al., 2024. Lawllm: Intelligent legal system with
694 legal reasoning and verifiable retrieval, in: International Conference on
695 Database Systems for Advanced Applications, Springer. pp. 304–321.

696 Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R.,
697 Smola, A.J., 2017. Deep sets, in: Advances in Neural Information Process-
698 ing Systems.

699 Zhang, T., Ramakrishnan, R., Livny, M., 1996. BIRCH: an efficient data
700 clustering method for very large databases, in: Proceedings of the 1996
701 ACM SIGMOD International Conference on Management of Data, pp. 103–
702 114.

703 Zhang, Y., Zhai, S., Meng, Y., Bi, S., Chen, Y., Qi, G., 2024. Event is
704 more valuable than you think: Improving the similar legal case retrieval
705 via event knowledge. *Information Processing & Management* 61, 103729.

706 Zhong, W., Guo, L., Gao, Q., Ye, H., Wang, Y., 2024. Memorybank: En-
707 hancing large language models with long-term memory, in: Proceedings of
708 the AAAI Conference on Artificial Intelligence, pp. 19724–19731.

709 **Appendix A. Numeric grids and controls**

710 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
 711 Soft O2 numeric grids appear below. Primary Soft O2 and Mix claims are
 712 carried by the main-text figures and tables.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: McNemar mid- p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE). Holm-adjusted p is reported for the nine primary Soft O2 vs FlatIP contrasts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	8.09e-47
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243	—
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	4.67e-17
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435	—
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982	—
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26	—
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05	—
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41	—
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302	—
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239	0.191
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10	—
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	6.36e-04
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035	—
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.68e-14
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09	—
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	8.75e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215	—
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	9.44e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04	—

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	0.554	[0.530,0.610]
	Soft O2	0.788	0.664	0.776	[0.757,0.820]
	Hard hydr.	0.802	0.578	0.864	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.411	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.445	[0.237,0.307]
	Soft O2	0.698	0.654	0.793	[0.660,0.735]
	Hard hydr.	0.720	0.585	0.847	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.369	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.420	[0.212,0.280]
	Soft O2	0.762	0.659	0.791	[0.725,0.797]
	Hard hydr.	0.790	0.586	0.854	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.348	[0.172,0.232]

713

Soft O2 leads EC; hard hydration can lead AH.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.640	0.741	0.730
	Soft O2	0.638	0.780	0.813
	Hard hydr.	0.614	0.598	0.886
	Shuffled O2	0.448	0.645	0.666
advice-recall	FlatIP	0.342	0.438	0.466
	Soft O2	0.428	0.503	0.524
	Hard hydr.	0.402	0.405	0.572
	Shuffled O2	0.310	0.421	0.436
u-param	FlatIP	0.547	0.681	0.681
	Soft O2	0.648	0.735	0.759
	Hard hydr.	0.592	0.571	0.827
	Shuffled O2	0.469	0.610	0.627

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.692	0.806	0.753
	Soft O2	0.752	0.848	0.869
	Hard hydr.	0.612	0.615	0.897
	Shuffled O2	0.570	0.747	0.705
advice-recall	FlatIP	0.332	0.464	0.483
	Soft O2	0.472	0.540	0.582
	Hard hydr.	0.394	0.403	0.601
	Shuffled O2	0.348	0.467	0.454
u-param	FlatIP	0.568	0.718	0.744
	Soft O2	0.759	0.819	0.844
	Hard hydr.	0.596	0.594	0.864
	Shuffled O2	0.592	0.712	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460